Criteria

In what follows, we focus on the problem of searching for Web pages. At the outset, we must define what kind of pages we're looking for when we conduct a search. The most important criterion, of course, is relevance. The page that comes up should contain something very similar to (if not exactly) what you are looking for. But relevance is not the only thing. If you're doing a search on crocodiles, it's very likely that you're also interested in alligators; therefore, ideally, a search for 'crocodile' should show up results that have only the word 'alligator' in them, in addition to those that contain 'crocodile'.

There's more. How authoritative is a page? You sure can't trust everything you see on the Internet. The page you're going to refer to should be, in some way, 'better' than other similar pages.

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A related criterion is popularity. You'd rather get your information from a popular source—assuming it is authoritative as well—than from an obscure source. Actually, these two are related, so we could just say we want pages that are authoritative and popular. Popular pages are more likely to be authoritative than unpopular ones for obvious reasons.

In sum, then, what is desired of the ideal search engine is that it displays not millions of pages that may or may not interest you, but a few authoritative, popular pages that address your need—in addition to other pages that may interest you, which you didn't think of when you typed in your search.

This is possible, firstly, if your search query is good; second, if the search engine determines your intent well; third, if the pages out there properly and honestly indicate to the search engine what they are about.

Now that's a lot of variables for a single search, and only if they come together every time you search will the results be of any use.

Good Search Queries

Technology is supposed to make life simpler. That includes making it unnecessary for us to rack our brains when we're conducting a search. There should be no need to be an 'expert Googler', if there were such a term. However, most of today's search engines actually do require that—that you frame your query well. If you're searching for a local place that serves up pizza, and you'd like to order online, you should be able to just type in 'pizza' and get your restaurant. Now that would entail some level of localisation capabilities built into the search engine, and also the fact that the engine should be able to determine what your intent is—namely, that you want to buy pizza.

Similarly, if you type in 'When was Britney born?', the search engine should be able to take that as a question and give you the answer. That also entails that the engine should be able to determine that you mean Britney Spears, not any other Britney out there. And as we mentioned earlier, returning popular results is important, and here, a popular page is much more likely to be about Britney Spears than about any other Britney.

At the time of going to print, a Google search on 'When was Britney born?' provides the answer, just as though what you typed in were a question. (Note that this happens when you use Google.com, not Google.co.in.) None of the other major search engines do, probably being thrown off by news of Britney Spears becoming a *born*-again Christian, and by the song *Born to make you happy*.

So is Google indeed God? It's premature to make such a conclusion, but the fact is that Google does, in some measure, seem to understand what you're looking for better than the other search engines do. In an ideal world, every search engine should be able to provide the answer to that question, just as if it were a question—and without the need for a specialised sub-engine that understands queries for dates!

How? The answer could just lie in AI (Artificial Intelligence), and the semantic mapping of the Web.

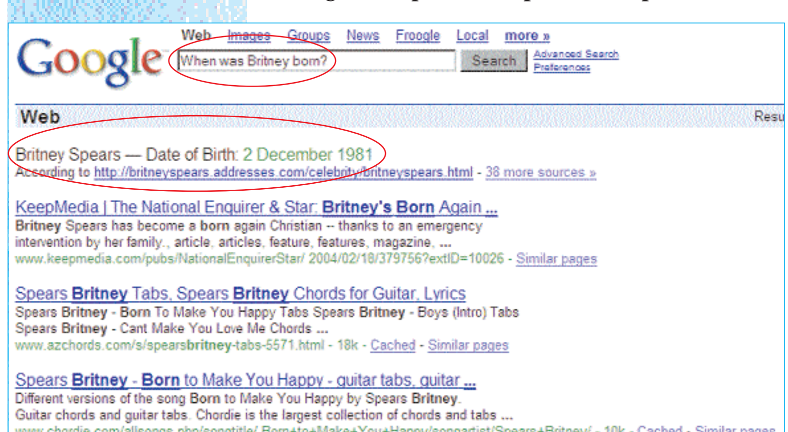
The Semantic Web?

If much of the problem with finding the right page lies in the information that the pages give out to search engine crawlers, the answer might just lie in the vision of Tim Berners-Lee called the Semantic Web, in the context of the WWW.

From Semanticweb.org, "the Semantic Web is a vision: the idea of having data on the web defined and linked in a way that it can be used by machines." It is, essentially, a project that aims to create a more intelligent Web by annotating pages on the Web with their semantics (meaning), in a manner understandable by computers (or search engine spiders). Thus, if a spider knew what a page was about, it would return more relevant results—or so the idea goes.

Let's take the Britney example again: there is no way for a spider or any other automated agent to know that Britney Spears is a person; that something like "2nd December 1981" is a date; or even that any given person has a special date called a birth date.

Using XML and other technologies, this information can be made explicit in the page that contains these elements. And what a great help that would be! If your search engine could understand things such as birthdates and people, and if pages could declare themselves as being descriptive of a particular person and a



Google is the only search engine that interprets a date-of-birth query as such, and gives the answer as a special result placed before the other results